

Data Encryption in Use

CompTIA SecAI+: AI Security Certification Complete Exam Prep 2026 | Section 13: Data Security Controls

1. The Memory Problem: Why Encryption in Use Exists

Encryption at rest protects stored data. Encryption in transit protects network-borne data. But for data to be processed — for an AI model to run inference, for a database to execute a query, for an application to render a web page — the data must be decrypted into CPU-accessible memory. This moment of decryption is the boundary at which the protection provided by at-rest and in-transit encryption ends.

For most applications and most data sensitivity levels, this gap is acceptable. Access controls, OS-level isolation, and network segmentation limit who can reach memory contents. But for specific high-sensitivity scenarios — medical AI processing identifiable patient data in a shared cloud environment, financial AI handling trade-secret algorithms, intelligence AI analysing classified feeds — the ability of a cloud provider, a hypervisor administrator, a sufficiently privileged insider, or a sophisticated attacker with kernel access to read memory contents creates an unacceptable residual risk. Encryption in use is the set of technologies developed to close this gap.

2. Confidential Computing: Hardware-Enforced Secure Enclaves

Confidential computing is an umbrella term for hardware technologies that create isolated execution environments — called trusted execution environments (TEEs) or secure enclaves — where code and data are protected by CPU hardware even from the operating system, the hypervisor, and in cloud deployments, the cloud provider's infrastructure team. The CPU hardware enforces the isolation: memory pages belonging to the enclave are encrypted by the processor itself, and the decryption keys are held within the CPU's secure boundary.

The three major processor architectures each implement TEEs with different technical characteristics but the same fundamental security model:

- Intel SGX (Software Guard Extensions): Creates enclaves of up to a few gigabytes of encrypted protected memory. Code and data inside the enclave are encrypted at the cache boundary; the CPU decrypts only when data enters CPU registers for computation. SGX is widely deployed in cloud platforms and supports remote attestation.
- AMD SEV (Secure Encrypted Virtualization): Encrypts entire virtual machine memory at the hardware level. Each VM receives a unique hardware-managed encryption key; the hypervisor and other VMs cannot read the encrypted memory. AMD SEV-SNP (Secure Nested Paging) extends this with integrity protection against memory tampering.
- ARM TrustZone: Partitions the processor into a 'normal world' and a 'secure world'; code running in the secure world is isolated from OS-level code. Widely used in mobile devices and IoT; also relevant for edge AI deployments.

Confidential computing vs. full disk encryption: Full disk encryption (FDE) protects data on storage media — once data is loaded into RAM, FDE provides no protection. Confidential computing extends protection into RAM and the compute layer itself. The two technologies are complementary, not alternatives.

3. Remote Attestation: Verifying the Enclave Before Trusting It

A TEE provides isolation guarantees, but a remote party needs a way to verify that they are actually communicating with a legitimate, unmodified enclave running the expected code — and not a compromised or simulated environment. Remote attestation is the mechanism that provides this assurance.

In the Intel SGX attestation flow, the enclave generates a cryptographic quote — a signed measurement of the enclave's contents — using a key that is derived from the Intel hardware itself. A remote party sends this quote to Intel's Attestation Service (or an equivalent third-party attestation service) which verifies the signature and confirms that the measurement matches the expected code. Only if attestation succeeds does the remote party send sensitive data into the enclave.

Attestation Step	What Happens	Security Purpose
1. Enclave measurement	CPU generates a cryptographic hash of enclave code and data layout	Establishes a tamper-evident identity for the specific code loaded
2. Quote generation	Enclave generates a signed attestation report using hardware-derived key	Binds the measurement to the specific, authentic CPU hardware
3. Quote verification	Remote party submits quote to attestation service	Third party confirms quote was signed by genuine Intel/AMD hardware
4. Secure channel setup	Remote party establishes encrypted channel to verified enclave	Ensures sensitive data flows only to the verified, trusted environment

4. Homomorphic Encryption: Computing on Ciphertext

Homomorphic encryption (HE) is a cryptographic technique that allows computations to be performed directly on encrypted data without decrypting it first. The result of the computation, when decrypted, is identical to the result that would have been obtained by performing the computation on unencrypted data. In effect, the server performing the computation never sees the plaintext — it manipulates mathematical representations of ciphertext that produce meaningful encrypted outputs.

Fully Homomorphic Encryption (FHE) supports arbitrary computations on ciphertext. Partial and Somewhat Homomorphic Encryption (PHE, SHE) support limited operation types (typically addition or multiplication but not both without restrictions). Current FHE implementations based on the CKKS scheme (Cheon-Kim-Kim-Song) support approximate real-number computations suitable for machine learning operations, but at a performance cost that makes real-time AI inference impractical as of 2026.

HE Type	Supported Operations	Performance	Practical AI Use (2026)
Partially Homomorphic (PHE)	Either addition OR multiplication (not both)	Moderate	Limited: private set intersection, simple

			aggregations
Somewhat Homomorphic (SHE)	Both operations up to a limited depth	Slow	Research: small-scale ML inference on encrypted data
Fully Homomorphic (FHE)	Arbitrary computation on ciphertext	Very slow (100x–1000x overhead)	Not yet practical for complex real-time AI inference

Real-World Deployment — Microsoft SEAL and Healthcare AI (2021–2024): Microsoft's SEAL library (Simple Encrypted Arithmetic Library) enabled a proof-of-concept deployment in which a healthcare organisation computed risk scores on encrypted patient data at a third-party cloud AI service. The service never decrypted the input data; only the healthcare organisation's systems could decrypt the results. While the computation took several minutes per patient record — making it impractical for real-time clinical decisions — it validated FHE's applicability for batch analytics on highly regulated data. This case established that FHE has a current deployment niche in batch, high-sensitivity analytics even while remaining impractical for real-time inference.

5. Secure Multi-Party Computation: Collaborative AI Without Data Sharing

Secure Multi-Party Computation (SMPC or MPC) is a cryptographic protocol family that allows multiple parties to jointly compute a function over their combined private inputs without any party revealing their input to any other party. The output — and only the output — is shared; each party's input remains confidential throughout the computation.

In AI contexts, the most significant application of MPC is collaborative model training across organisations that cannot legally or competitively share their raw data. Consider three competing hospitals that collectively have enough oncology imaging data to train a high-accuracy AI diagnostic model, but each is legally prohibited from sharing patient data with the others. MPC enables all three to jointly train the model without any party receiving access to the others' imaging data.

- Secret sharing schemes: Each party's input is split into shares distributed across all participants. No individual party holds enough shares to reconstruct any other party's input, but all parties together can evaluate the function.
- Garbled circuits: A technique that represents the computation as a Boolean circuit and 'garbles' it so that computation can proceed on encrypted inputs without revealing intermediate values.
- Oblivious transfer (OT): A sub-protocol used within MPC that allows one party to receive exactly one of several possible pieces of information without the sender knowing which was received — a fundamental building block for many MPC schemes.
- Performance characteristics: MPC introduces communication overhead proportional to the number of parties and circuit complexity. For large-scale ML, MPC is typically combined with other privacy-preserving techniques rather than used in isolation.

6. Federated Learning: Privacy-Preserving Training Without Centralisation

Federated learning is a distributed machine learning paradigm in which training occurs locally on data held by each participant, and only model updates (gradients) — not raw data — are shared with a central aggregation server. The central server never has access to any participant's training data; it sees only the numerical gradient updates that describe how each local model changed during its training epoch.

Federated learning provides privacy benefits but is not, by itself, a cryptographic privacy guarantee. Gradient inversion attacks have demonstrated that it is theoretically possible to reconstruct training data from gradient updates under certain conditions. Secure aggregation — an MPC-based protocol that aggregates gradients cryptographically so the server sees only the sum, not individual updates — is the cryptographic complement to federated learning that closes this gap.

Technique	What Is Shared	Central Server Visibility	Relevant Threat Mitigated
Standard (centralised) training	Raw training data	Full plaintext data access	None — baseline; all others improve on this
Federated learning (basic)	Model gradients only	Gradients (not raw data)	Data minimisation; reduces direct data exposure
Federated + secure aggregation	Encrypted gradient shares	Aggregate sum only; no individual updates	Gradient inversion attacks; insider threat at aggregator
Federated + differential privacy	Noisy gradients	Aggregate sum with statistical noise	Membership inference attacks; individual record reconstruction

7. Memory Encryption at the Hardware Level

Beyond secure enclaves, some processor architectures provide system-wide RAM encryption that protects all memory contents rather than only enclave regions. AMD Secure Memory Encryption (SME) and AMD Transparent Secure Memory Encryption (TSME) encrypt all data written to DRAM using a key generated by the processor during boot. This provides protection against cold boot attacks — where an attacker with physical access quickly freezes and removes memory modules to read their contents before the DRAM charge decays.

For cloud AI deployments, AMD SEV extends this concept to encrypt each virtual machine's memory with a separate key managed by the AMD Secure Processor, invisible to the hypervisor. Azure Confidential VMs and Google Confidential VMs use AMD SEV to provide memory encryption for cloud-hosted AI workloads. Intel's equivalent, Total Memory Encryption (TME), provides system-wide DRAM encryption and is the technology underlying some Intel-based confidential computing offerings.

8. Performance Trade-offs and Deployment Decision Framework

No encryption-in-use technology is free from performance impact, and the magnitude of impact varies substantially across approaches. Understanding these trade-offs is essential for selecting the right technology for a given AI security requirement.

Technology	Approx. Overhead	Deployment Complexity	Best Suited For
Intel SGX enclave	10–30% CPU	Moderate; application	Small, high-sensitivity AI tasks;

	overhead; limited enclave memory	may need partitioning	key management; attestation-required scenarios
AMD SEV (full VM)	5–15% CPU overhead; VM-wide protection	Low; transparent to applications	Lifting cloud AI VMs into confidential computing
FHE (CKKS scheme)	100x–1000x slower than plaintext computation	High; requires domain expertise	Batch analytics on highly regulated data; not real-time inference
SMPC	Significant communication overhead; scales with parties	High; multi-party protocol management	Collaborative model training across untrusting organisations
Federated learning	Bandwidth for gradient exchange; local compute	Moderate; orchestration infrastructure	Privacy-preserving distributed training at scale

The decision framework for selecting an encryption-in-use approach should begin with the question: what is the specific threat being mitigated? If the threat is a cloud provider with potential physical access to hardware, AMD SEV addresses it efficiently. If the threat is a third-party computation service that must not see plaintext inputs, FHE or TEE-based attestation is appropriate. If the threat is exposing raw training data in a multi-institutional collaboration, federated learning with secure aggregation is the architecturally appropriate choice.

9. Analogy: Encryption in Use as Conducting Surgery in a Sealed Clean Room

Encryption at rest is like locking patient files in a cabinet. Encryption in transit is like transporting them in an armoured vehicle. But when a surgeon needs to read those files during an operation, they must be taken out of the locked cabinet — exposing them to anyone in the operating room. Confidential computing is the equivalent of building a sealed clean room where the files can be consulted, with hardware-enforced walls that prevent anyone outside the room — even hospital administrators — from seeing what is inside. The surgical team (the computation) operates inside the sealed environment; the results come out, but the files themselves never leave the room in readable form.

Homomorphic encryption extends the analogy: it is as if the surgeon could perform the diagnosis without ever opening the file — working through opaque envelopes to produce a result, with no one ever seeing the original documents. The result is accurate, but the process is far slower than working with open files. This performance cost is why homomorphic encryption is reserved for scenarios where the privacy requirement is so strong that slower computation is an acceptable trade-off.

Key Terms Glossary

Term	Definition
AMD SEV (Secure Encrypted Virtualization)	AMD processor feature that encrypts individual virtual machine memory with per-VM keys, preventing the hypervisor and other VMs from reading encrypted memory.
Attestation (remote)	A cryptographic mechanism by which a TEE proves to a remote

	party that it is running specific, unmodified code on genuine hardware, before sensitive data is sent.
Confidential computing	Hardware-based technologies that create trusted execution environments protecting data during computation from the OS, hypervisor, and cloud provider.
Federated learning	A distributed ML training approach where participants train locally and share only model updates (gradients), not raw data, with a central aggregation server.
Fully Homomorphic Encryption (FHE)	A cryptographic scheme enabling arbitrary computation on encrypted data without decryption; results decrypt to match plaintext computation results.
Intel SGX (Software Guard Extensions)	Intel CPU feature creating encrypted memory enclaves protected from the OS and hypervisor; supports remote attestation.
Secure Enclave	A hardware-protected region of CPU memory where code and data are encrypted by the processor itself; only accessible to the designated application inside the enclave.
Secure Multi-Party Computation (SMPC)	Cryptographic protocols enabling multiple parties to jointly compute a function over private inputs without any party revealing their input to others.
Trusted Execution Environment (TEE)	An isolated, hardware-protected execution context that provides confidentiality and integrity for code and data running within it.

Quick Revision Checklist

- Explain the memory problem: encryption at rest and in transit do not protect data loaded into RAM for processing.
- Name the three major TEE implementations: Intel SGX, AMD SEV, and ARM TrustZone.
- Describe remote attestation and its role in confidential computing deployments.
- Distinguish FHE (arbitrary computation on ciphertext) from PHE (limited operations only).
- State the current practical limitation of FHE: 100x to 1000x performance overhead makes real-time AI inference impractical as of 2026.
- Explain secure multi-party computation and identify its primary AI use case: collaborative training across untrusting organisations.
- Describe federated learning and the gradient inversion attack that motivates secure aggregation.
- Know the performance overhead ranges: AMD SEV 5 to 15 percent; Intel SGX 10 to 30 percent; FHE 100x to 1000x.
- Identify when to recommend confidential computing versus FHE versus federated learning based on the specific threat being mitigated.
- Explain the cold boot attack and identify which memory encryption technology (AMD SME/TSME) addresses it.

10 Exam-Based MCQs — Data Encryption in Use

Test yourself after reading. These questions are written in real exam style — scenario-heavy, with plausible distractors. Read every explanation, including for questions you answered correctly.

Q1. Scenario: *The cloud provider's terms of service acknowledge that infrastructure staff may access virtual machine memory for operational troubleshooting. The patient data is identifiable and the organisation is subject to HIPAA.* A healthcare AI company processes identifiable patient records using a large language model deployed on a major cloud provider's infrastructure. The CISO is concerned that the cloud provider's infrastructure team could theoretically access patient data in memory during inference. Which technology MOST directly addresses this specific threat?

- A) TLS 1.3 encryption for all API calls to the inference service
- B) AES-256 encryption of the patient records in the database
- C) Confidential computing using AMD SEV or Intel SGX to protect the inference workload in a hardware-isolated TEE
- D) Role-based access controls limiting cloud provider staff from accessing the company's account

Answer: C **Explanation:** C is correct. The specific threat is the cloud provider's infrastructure team reading data from VM memory — a capability that operating-system-level access controls and IAM policies cannot prevent, since the hypervisor layer sits above the guest OS. Confidential computing (AMD SEV or Intel SGX) encrypts VM memory at the hardware level with keys managed by the CPU's secure processor; the hypervisor and cloud provider staff cannot access the decrypted memory contents. A (TLS) protects data in transit to the inference service but does not protect it during processing. B (AES-256 at rest) protects stored data but data must be decrypted for inference. D (RBAC) operates at the account control plane level and does not prevent infrastructure-level memory access.

Q2. Scenario: *The collaboration agreement explicitly prohibits any party from receiving another party's raw clinical trial data. Legal counsel has confirmed that regulatory requirements in their jurisdictions reinforce this prohibition.* Three competing pharmaceutical companies want to collaboratively train an AI drug interaction model using each company's proprietary clinical trial data. None of the companies is willing to share its raw trial data with the others. Which technology architecture MOST directly enables this collaboration without exposing any company's raw data to the others?

- A) Centralised training on anonymised data contributed by all three companies
- B) Secure multi-party computation (SMPC) allowing joint training on encrypted inputs without revealing individual company data
- C) Training three separate models and averaging their parameters
- D) Using a third-party cloud provider as a trusted neutral aggregation point

Answer: B **Explanation:** B is correct. SMPC protocols allow multiple parties to compute a joint function — in this case, model training — over their combined private inputs without any party's raw data being revealed to any other party or to the central aggregation infrastructure. A requires data sharing (even if anonymised) which the scenario prohibits. C (ensemble averaging) avoids data sharing but does not produce a collaborative model; it produces independent models with averaged parameters, which is a different and weaker approach. D introduces a trusted third party — but the scenario implies the companies do not fully trust each other or a neutral third party, and the cloud provider could still be subpoenaed or breached.

Q3. Scenario: *The vendor will process customer purchase history and browsing data to generate product recommendations displayed in real time as the customer browses the company's e-commerce platform.*

A security engineer is evaluating whether to use Fully Homomorphic Encryption (FHE) to allow a third-party AI vendor to run inference on the company's proprietary customer data without seeing it. The AI model performs real-time product recommendations with a target response time of under 500 milliseconds. What is the PRIMARY limitation of FHE for this use case?

- A) FHE does not support the mathematical operations required for neural network inference
- B) FHE is impractical for real-time inference because current implementations add 100 to 1000 times the computation overhead compared to plaintext processing
- C) FHE requires the vendor to hold the customer's encryption key, defeating the privacy purpose
- D) FHE is only approved for government use and cannot be used in commercial AI applications

Answer: B **Explanation:** B is correct. The fundamental practical limitation of FHE as of 2026 is computational overhead. Even the most optimised FHE schemes (such as CKKS for approximate arithmetic) are orders of magnitude slower than plaintext operations for ML inference workloads. A 500-millisecond target is incompatible with current FHE performance for neural network inference. A is incorrect; CKKS and related schemes do support the operations in neural network inference (multiply-accumulate), but the performance cost is prohibitive. C is incorrect; FHE does not require key sharing — the client encrypts data with their key, the server computes on ciphertext, and the client decrypts the result. D is incorrect; FHE is available in open-source libraries (Microsoft SEAL, Google's Concrete ML) for commercial use.

Q4. A data scientist working with federated learning reports that gradient updates shared by remote participants can theoretically be used to reconstruct individual training records — a technique known as gradient inversion. Which complementary technology MOST directly mitigates this risk?

- A) Encrypting the training dataset at rest on each participant's device
- B) Secure aggregation — an MPC-based protocol that allows the aggregation server to see only the sum of gradients, not individual updates
- C) Using a larger batch size during local training to make gradient updates less informative
- D) Replacing gradient sharing with sharing only model weights at the end of training

Answer: B **Explanation:** B is correct. Secure aggregation is the cryptographic solution specifically designed to complement federated learning by preventing the aggregation server from seeing individual gradient updates. Using an MPC-based protocol, participants encrypt their gradient updates such that the server can compute only the aggregate sum — which is needed for model updates — without accessing any individual participant's contribution. Gradient inversion requires access to individual gradients, not the aggregate. A (at-rest encryption) does not affect gradient sharing. C (larger batch size) reduces gradient information density but does not eliminate reconstruction risk and is an operational approximation, not a cryptographic guarantee. D describes a different (and less useful) information-sharing approach that does not solve gradient inversion.

Q5. Which statement MOST accurately describes the relationship between Intel SGX's remote attestation capability and the privacy guarantee provided by the enclave?

- A) Remote attestation is optional and only required when regulatory compliance mandates it
- B) Remote attestation allows a party sending sensitive data to verify that the enclave is running specific, unmodified code on genuine Intel hardware before transmitting data
- C) Remote attestation replaces the need for encryption inside the enclave because the code integrity guarantee ensures safe data handling
- D) Remote attestation is performed by the cloud provider, who uses it to verify that their customers are running approved applications

Answer: B **Explanation:** B is correct. The privacy guarantee of a TEE is meaningless if the remote party cannot verify that the enclave is actually running the expected code on genuine hardware — without attestation, a malicious actor could simulate an enclave environment to steal data. Remote attestation closes this gap by providing a cryptographically verifiable proof of enclave identity. A is incorrect; attestation is architecturally essential for TEE-based privacy, not optional. C is incorrect; attestation and encryption are complementary — attestation verifies identity, encryption protects data in transit to the verified enclave. D is incorrect; attestation is performed by the enclave itself against Intel's (or AMD's) attestation service, not by the cloud provider.

Q6. An AI security architect is comparing AMD SEV and Intel SGX for protecting a large language model inference workload in a cloud environment. The model requires 40 GB of RAM for inference operations. Which factor is MOST likely to make Intel SGX impractical for this specific workload?

- A) Intel SGX does not support the AVX-512 instructions required for efficient matrix multiplication
- B) Intel SGX enclave memory is limited to a few gigabytes in current implementations, making it insufficient for a 40 GB LLM inference workload
- C) Intel SGX requires a dedicated physical server and cannot be used in virtualised cloud environments
- D) Intel SGX adds 50 percent overhead to all RAM operations, making it twice as slow as AMD SEV for memory-intensive workloads

Answer: B **Explanation:** B is correct. Intel SGX enclave memory (the Enclave Page Cache, or EPC) is limited — practical limits in current deployments range from a few hundred MB to a few GB, far below the 40 GB required for the workload described. AMD SEV, by contrast, encrypts the entire VM's memory without the size constraints of SGX enclaves, making it better suited for large-scale AI workloads. A is incorrect; SGX's memory size limitation, not instruction set support, is the binding constraint. C is incorrect; SGX is widely available in virtualised cloud environments (Azure DCsv2/DCsv3 series, for example). D overstates the overhead differential between the two technologies.

Q7. Google uses federated learning for keyboard predictions on Android devices, where each user's typing data remains on their phone and only model updates are shared. Which privacy property does this architecture PRIMARILY provide?

- A) Encryption in use — the model computations occur inside a secure enclave on the Android device

- B) Data minimisation — raw typing data never leaves the device; only gradient updates are transmitted
- C) End-to-end encryption — all data is encrypted from the user's device to Google's servers
- D) Anonymisation — the model updates are stripped of any personally identifiable information before transmission

Answer: B **Explanation:** B is correct. The primary privacy property of federated learning is data minimisation — the sensitive raw data (in this case, typing patterns and content) never leaves the originating device. Only mathematical model updates (gradients) are transmitted, reducing both the data exposure surface and the regulatory risk. A is incorrect; standard federated learning does not require secure enclaves on participant devices, though this can be combined. C is incorrect; while TLS would be used for transmission, E2EE from a privacy standpoint requires that the server cannot access the plaintext — in basic federated learning, gradients may be visible to the aggregation server. D is incorrect; gradient updates are not anonymised data; they are mathematical derivatives that can, under certain conditions, leak information about the underlying training data.

Q8. Scenario: The attacker obtained the server from an asset disposal company that acquired decommissioned hardware. The server's SSDs were wiped but the RAM was not considered. An attacker with physical access to a decommissioned cloud server quickly freezes the server's RAM modules with liquid nitrogen and removes them, reading the DRAM contents after power removal (a cold boot attack). Which memory-level technology MOST directly mitigates this specific attack?

- A) Full disk encryption (BitLocker or LUKS) on the server's storage volumes
- B) AMD Secure Memory Encryption (SME) or Intel Total Memory Encryption (TME), which encrypt DRAM contents at the hardware level
- C) Enabling Secure Boot on the server to prevent unauthorized OS loading
- D) Disabling hibernation and sleep states to prevent memory content persistence

Answer: B **Explanation:** B is correct. AMD SME and Intel TME encrypt DRAM contents using keys generated by the processor, stored within the CPU. Even if DRAM modules are removed from the server, their contents are ciphertext. Cold boot attacks rely on reading DRAM contents directly after removal — memory encryption renders this unreadable without the CPU-held key. A (FDE) protects storage media, not RAM. C (Secure Boot) prevents loading unauthorised operating systems but does not protect DRAM contents from physical extraction. D (disabling sleep/hibernate) reduces the window for software-based memory attacks but does not address hardware extraction of DRAM modules.

Q9. Scenario: The API passes through an AWS Application Load Balancer, a Kong API gateway, and two microservices before reaching the inference engine. TLS 1.3 is terminated and re-established at each hop. A compliance officer asks an AI security architect to explain the difference between 'encryption in transit' and 'end-to-end encryption' in the context of an AI inference API. The current implementation uses TLS 1.3 for all API traffic and AES-256 for data at rest. Which response MOST accurately characterises the gap?

- A) There is no difference — TLS 1.3 provides end-to-end encryption for all traffic to the API

- B) TLS 1.3 encrypts traffic between each hop; at each intermediate service, data is decrypted and re-encrypted. True E2EE would require data to remain encrypted through all intermediate processing, achievable via confidential computing or FHE
- C) End-to-end encryption applies only to messaging applications and is not a relevant concept for AI inference APIs
- D) The gap can be closed by upgrading from TLS 1.3 to TLS 1.4 when it becomes available

Answer: B **Explanation:** B is correct. TLS encrypts each individual hop in a communication chain — when traffic reaches an intermediate service (a load balancer, an API gateway, a preprocessing layer), TLS terminates and the service processes plaintext before establishing a new TLS session to the next hop. E2EE in an AI inference context means the data remains encrypted even while being processed by the inference service — which requires confidential computing (the service processes data inside a TEE) or FHE (the service computes on ciphertext). A is incorrect for this reason. C is incorrect; E2EE is a well-defined concept applicable to AI systems. D is incorrect; TLS version changes address cipher suites and protocol security, not the fundamental hop-by-hop decryption architecture.

Q10. An AI security team is designing a privacy-preserving analytics system where multiple hospitals contribute patient data to train a cancer detection model. The requirements are: (1) no hospital's raw data leaves its premises, (2) the aggregation server must not see individual hospital gradient updates, (3) the final model must be as accurate as if trained on the combined dataset. Which combination of technologies BEST satisfies all three requirements?

- A) Centralised training with differential privacy applied to the final model
- B) Federated learning with secure aggregation — local training at each hospital; secure MPC-based aggregation that reveals only the sum of gradients to the aggregation server
- C) Homomorphic encryption — each hospital encrypts its dataset and sends it to a central server for training
- D) Three independent models trained at each hospital, with model outputs averaged at inference time

Answer: B **Explanation:** B is correct. Federated learning satisfies requirement (1): raw data never leaves each hospital's infrastructure, only gradients do. Secure aggregation satisfies requirement (2): the MPC-based aggregation protocol ensures the server sees only the aggregate gradient sum, not individual hospital contributions. The resulting model is trained on the effective union of all hospitals' data (through gradient aggregation), satisfying requirement (3). A violates requirement (1) — centralised training requires data to be moved. C uses FHE for the wrong purpose — FHE enables computation on encrypted data at a server, but the data still travels to the server (requirement 1 is violated). D satisfies requirement (1) but produces independent models, not a jointly-trained model, and the ensemble approach (ensemble averaging at inference) is significantly weaker than collaborative training for this use case.